

Mahima Rudrapati

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Education

University of California, Davis	Davis, CA
<i>Master of Science in Computer Science (Thesis), GPA: 3.91/4.00</i>	Sep 2024 – Dec 2026
University of Mumbai	Mumbai, India
<i>Bachelor of Engineering in Computer Engineering (Minor: Artificial Intelligence), GPA: 3.85/4.00</i>	Jul 2020 – May 2024

Technologies

3D Perception & Robotics: NVIDIA FoundationPose, Open3D, NeRF, 3DGS, 3D Reconstruction, ICP, Camera calibration
Languages: Python, C/C++, TypeScript/JavaScript, HTML/CSS3, SQL, NoSQL
Frameworks: React, React Native, Node.js, Flask, FastAPI, Django, Tailwind CSS
Libraries/ML: PyTorch, TensorFlow, OpenCV, NumPy, scikit-learn, multimedia libraries, SegmentAnything(SAM)
Databases & Tools: PostgreSQL, MongoDB, MySQL, Firebase, Git/GitHub, Linux, Docker, Figma

Experience

Graduate Student Researcher	Feb 2025 - Present
Center for Spaceflight Research, UC Davis	Davis, CA
<i>Advised by Dr. Stephen Robinson Research Focus: 3D Perception, Robotics, Autonomous Systems</i>	
- Architected an end-to-end 3D perception pipeline in Python for autonomous agriculture, collaborating with a cross-functional team of agriculturists and engineers to advance NASA's long-duration space habitat research. - Achieved under 2 cm error with my canopy height estimation algorithm for PAR-light sensing across crop beds using AprilTags, depth back-projection, SAM/SAM-3 segmentation and coordinate geometry. - Estimated hydroponic system localization with over 90% overlap using PyTorch, ICP, FoundationPose, and Open3D. - Built a 3D plant reconstruction pipeline using plant filtering and point cloud cleaning, feature matching using SIFT, and stitching using SFM enabling per-leaf segmentation and growth tracking over time. - Resolved multi-sensor fusion issues like RGB-depth alignment and calibration across Intel RealSense and Orbbec Femto Bolt sensors using custom intrinsics and testing algorithms.	
Software Engineer	Jun 2025 - Sep 2025
Pilotcrew AI (LLM and Data Evaluation Startup)	Davis, CA
- Built end-to-end LLM evaluation infrastructure, deployed on AWS, enabling real-time progress tracking and expert human evaluation across 20+ LLMs on custom datasets, with a cross-functional team of 5. - Shipped a file-chunking upload feature supporting 1GB+ datasets in under 75 seconds, with resume-on-failure, validation, and preprocessing, enabling reliable pilot testing and client access within 6 weeks.	
Teaching Assistant	Sep 2025 - Present
University of California, Davis	Davis, CA
Assisted in teaching C++, Data Structures, OOP, Javascript and Ubuntu to 100+ students through lectures, live coding and doubt clarification, receiving positive feedback for simplifying complex concepts and debugging errors.	

Publications

Restaurant Recommendation System using ML Algorithms and Real-Time Web IEEE	Nov 2024
DOI: 10.1109/DABCon63472.2024.10919373 Venue: Kolkata, India	
- Led a team of 4 to build a web-based restaurant recommendation platform using React, PostgreSQL, Flask and Python. - Integrated NLP-based sentiment analysis with content based and collaborative filtering based algorithms. - Improved latency of real-time data scraping pipelines by 75%, significantly improving data fetch and storage speeds compared to standard Selenium workflows. - Designed and implemented REST APIs and database schemas for large dataset storage of over 10K+ restaurant and review records.	

Projects

Realign: LLM Fine-Tuning and Benchmarking	github.com/blanklavender/ReAlign-Math
- Fine-tuned DeepSeek-Math-7B using LoRA and QLoRA on a custom dataset, managing a multi-day training run on remote GPU lab infrastructure. - Built an unsupervised benchmarking framework using semantic embeddings and dynamic programming to evaluate model reasoning quality beyond standard accuracy metrics.	